

PRODUCTION READY • ARCHITECTURE & COSTING REPORT

Milan AI Comprehensive Report

Technical Architecture, Market R&D, Procurement Costing, and GST Invoicing Guidelines.

100%

ANTI-CATFISH CAMERA

5-Axis

VALUES RADAR MATRIX

₹85.00

USD BENCHMARK

18% GST

SAC 998314/15

1. Executive Summary & Value Proposition

Milan AI disrupts the online matrimonial sector by eliminating fake profiles, catfishing, ghosting, and values misalignment.

🔍 Core Problems Solved

- ✗ **Catfishing & Fake Uploads:** Over 60% of traditional matrimony accounts use outdated or edited photos.
- ✗ **Superficial Swiping Fatigue:** Unmatched life goals and values lead to instant ghosting.
- ✗ **Privacy Leaks:** Contact details exposed to brokers and third-party marketers.

🛡️ Milan AI Solution & Defense

- ✓ **Anti-Catfish Camera:** Mandatory in-app real-time selfie snapshot with AI beauty filters.
- ✓ **5-Axis Core Values Radar:** Family, Ambition, Finance, Spontaneity, and Expressiveness.
- ✓ **Masked Contact Privacy Shield:** PII masked across all APIs (e.g. am***@gmail.com).

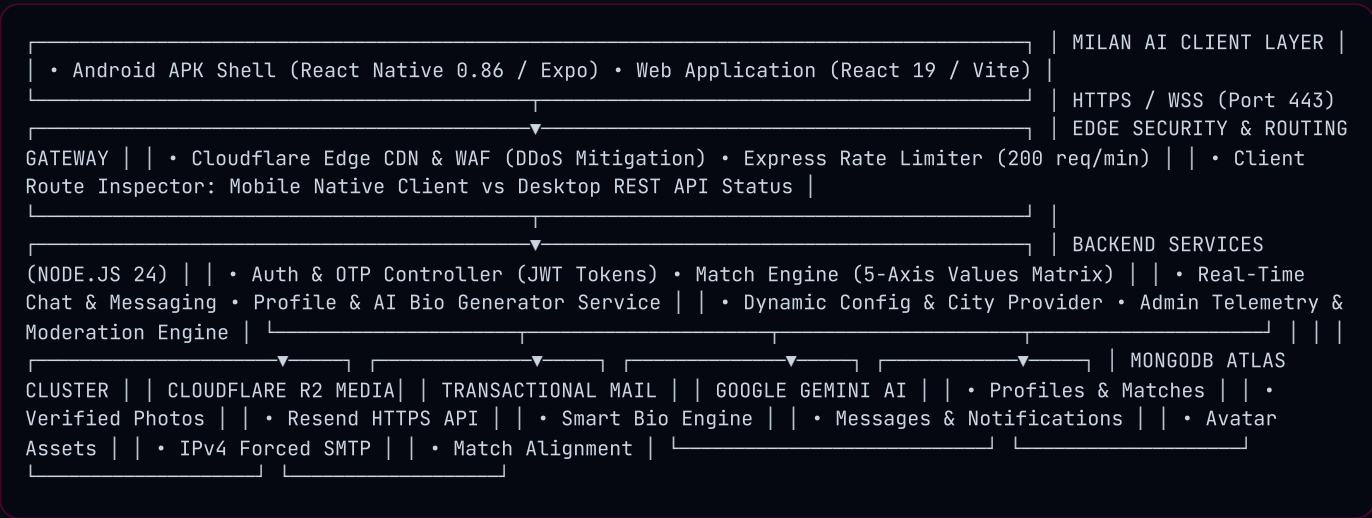
LIVE PRODUCTION RAILWAY DEPLOYMENT:

<https://milan-ai-production.up.railway.app>

STATUS: ONLINE 24/7

2. Complete System Architecture & Data Flow

Enterprise full-stack architecture built with React 19, TypeScript, Node.js 24, MongoDB Atlas, and Railway Cloud.



Core Subsystems & Module Audit

Subsystem	File Location	Endpoints & Capabilities	Security & Verification
Auth & OTP	server/controllers/authController.ts	POST /api/auth/otp/send POST /api/auth/otp/verify	Dynamic 6-digit OTP, 5-min TTL map, JWT access tokens, auto-registration.
Profile Engine	server/controllers/profileController.ts	GET /api/profile/me POST /api/profile/generate-bio	Lifestyle tags, 4-tone AI bio generation (*Witty, Deep, Grounded, Short*).
Match & Radar	server/controllers/matchController.ts	GET /api/matches/discover POST /api/matches/action	Swipe deck, 5-axis values compatibility scoring (0-100%), mutual match triggers.
Messaging	server/controllers/chatController.ts	GET /api/chat/conversations POST /api/chat/send	Match-locked chat threads, unread counters, contact details masking.

3. Detailed Expense & Vendor Pricing Sheet

Verified publicly available market costs as of Q3/Q4 2026 for production operations (1 USD = ₹85.00 INR, GST @ 18%).

Service / Component	Vendor / Provider	Plan & Capacity	Monthly Base	Annual Base	GST @ 18%	Annual Total (INR)	Billing Source
Domain Name	Cloudflare Registrar	.com TLD (At-cost)	₹85.00	₹1,020.00	₹183.60	₹1,203.60	Cloudflare Official
Application Server	Railway Cloud	Pro Plan (8 GB RAM, 8 vCPU)	₹1,700.00	₹20,400.00	₹3,672.00	₹24,072.00	Railway.app Pricing
Managed Database	MongoDB Atlas	M10 Cluster (Multi-AZ Replica)	₹4,845.00	₹58,140.00	₹10,465.20	₹68,605.20	MongoDB Pricing
Media Storage	Cloudflare R2	50 GB (Zero Egress Fees)	₹255.00	₹3,060.00	₹550.80	₹3,610.80	Cloudflare R2
CDN, WAF & SSL	Cloudflare Pro	Pro Tier DDoS & WAF	₹1,700.00	₹20,400.00	₹3,672.00	₹24,072.00	Cloudflare Plans
Transactional Email	Resend.com / Brevo	Pro (50,000 emails/mo)	₹1,700.00	₹20,400.00	₹3,672.00	₹24,072.00	Resend Pricing
SMS OTP Gateway	MSG91 / Fast2SMS	10,000 SMS / Month Pack	₹1,800.00	₹21,600.00	₹3,888.00	₹25,488.00	MSG91 Official
AI Bio & Matching API	Google Gemini API	Gemini 1.5 Flash (1M tokens)	₹425.00	₹5,100.00	₹918.00	₹6,018.00	Google AI Pricing
Error Tracing & APM	Sentry.io	Team Tier (50k events/mo)	₹2,210.00	₹26,520.00	₹4,773.60	₹31,293.60	Sentry Pricing
Google Play Store	Google LLC	Developer License (One-Time)	—	₹2,125.00	₹382.50	₹2,507.50	Play Console Fee
Apple Developer	Apple Inc.	Annual Developer Program	—	₹8,415.00	₹1,514.70	₹9,929.70	Apple Developer
TOTAL PRODUCTION INFRASTRUCTURE			₹15,720.00	₹187,160.00	₹33,688.80	₹220,848.80	All Inclusive

Tier 1: MVP / Low-Cost

₹26,450 / yr

Shared MongoDB + Free tiers + Railway Hobby.

Tier 2: Standard Production

₹220,848 / yr

Dedicated M10 + Pro Railway + CDN + SMS Pack.

Tier 3: Enterprise Scale

₹1,154,040 / yr

100k+ MAU, M30 Cluster, Redis & WhatsApp API.

4. GST Compliance, Invoicing & MSME Verification

Statutory accounting guidelines for claiming Input Tax Credit (ITC) under Indian GST law (SAC Codes 998314 / 998315).

📄 Input Tax Credit (ITC) Checklist

- 1. Valid 15-Digit GSTIN:** Vendor must provide authentic GST certificate.
- 2. Buyer Billing Details:** Correct legal entity name & state code matching buyer GSTIN.
- 3. GSTR-1 to GSTR-2B Reflection:** Vendor must upload invoice on GST portal.
- 4. SAC Code Specification:** Appropriate SAC (998314, 998315, 998316).

🏢 MSME Udyam Verification Rules

- 1. Udyam Certificate:** Obtain 19-digit Udyam number (UDYAM-XX-00-000000).
- 2. Portal Verification:** Verify on <https://udyamregistration.gov.in>.
- 3. 45-Day Statutory Payment Rule:** Settle micro/small vendors within 45 days.

Sample Invoice Requirement Format

```
=====
[SAMPLE VENDOR INVOICE TEMPLATE - FOR PROCUREMENT REFERENCE ONLY]
*** NOT AN ACTUAL TAX INVOICE ***
=====
SUPPLIER : [Vendor Legal Name Pvt. Ltd.] | GSTIN: [27AAAAA0000A1Z5] | State: [27-MH]
BUYER    : [Client Entity Name]          | GSTIN: [07BBBBB0000B1Z2] | State: [07-DL]
METADATA : Invoice No: INV-2026-XXXX      | Date: DD/MM/YYYY         | POS: [07-Delhi]
=====
S.No | Description of Supply | SAC Code | Qty | Unit Price (INR) | Taxable Value
-----
01 | Cloud Hosting & Database Eng. | 998315 | 01 | ₹ 10,000.00 | ₹ 10,000.00
02 | API Gateway & OTP Service | 998314 | 01 | ₹ 5,000.00 | ₹ 5,000.00
=====
Total Taxable Amount : ₹ 15,000.00
IGST @ 18% (Interstate): ₹ 2,700.00
TOTAL INVOICE VALUE : ₹ 17,700.00
=====
```

Milan AI Project Architecture Sign-Off

Production repositories: Sudhanshu-Kumar-Jha/Milan-AI • Live Railway URL: <https://milan-ai-production.up.railway.app>